Power Electronic Systems: Wireless Power Transfer

Author: John Poirier

Lab Dates: March 3 – June 6, 2025

Report Date: June 13, 2025

1. Introduction/Theory

The goal of this lab was to create a functional prototype wireless power transfer (WPT) board. This board was made modularly, so that each module could be made independently and placed into the overall circuit for testing and functionality. The prototype is made up of six separate modules: the Oscillator Module, the Charge Dump Module, the Switching Device Module, the Transformer Module, the Base Board Module, and the Receiver Module. The functionality and design of each of these modules is explained in detail in the next section of this report. Two experiments were conducted to test the functionality and efficiency of the WPT prototype. The first experiment tests the functionality of the Switching Device Modules and the Charge Dump Modules, and more specifically to test and compare Field Effect Transistors, and the second experiment examines the efficiency of the WPT prototype with different spacers between the transmitting coil and the receiving coil. These experiments and their results are explained more in the Results section of this report.

The method of wireless power transfer used in this lab is Inductive Wireless Power Transfer, which utilizes inductors, coils of wire, and magnetic fields generated between those coils. Changing magnetic fields induces an electromotive force on conductors, and two coils are used, a transmitting coil and a receiving coil, to transfer power through these electromotive forces.

2. Modular Prototype Development

The construction of this prototype was done modularly, so that testing could be conducted on each module individually, and as a whole. The complete prototype, with each module labelled, is shown below in **Figures 1 and 2**.

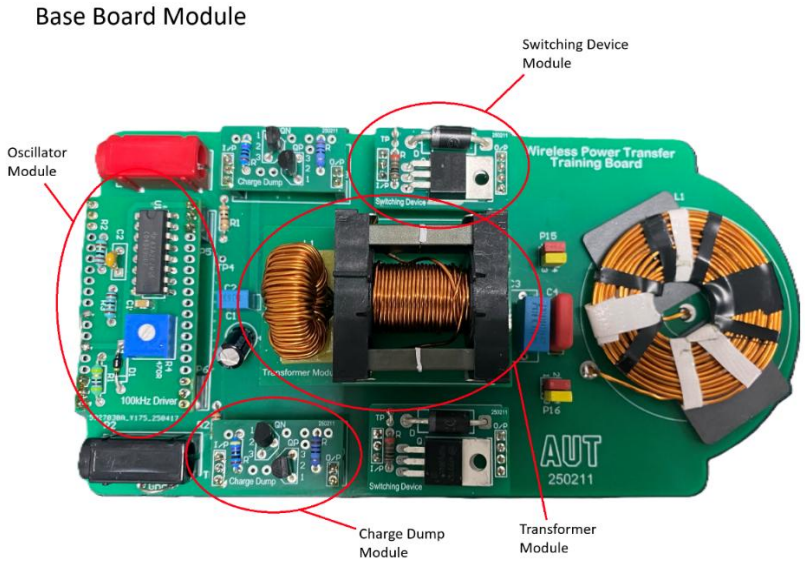


Figure 1: WPT Base Board with Modules Attached

Receiver Board Module



Figure 2: WPT Receiver Board Module

a) Oscillator Module

The first module constructed in this lab is the Oscillator Module. The schematic for this module is shown below in **Figure 3** below.

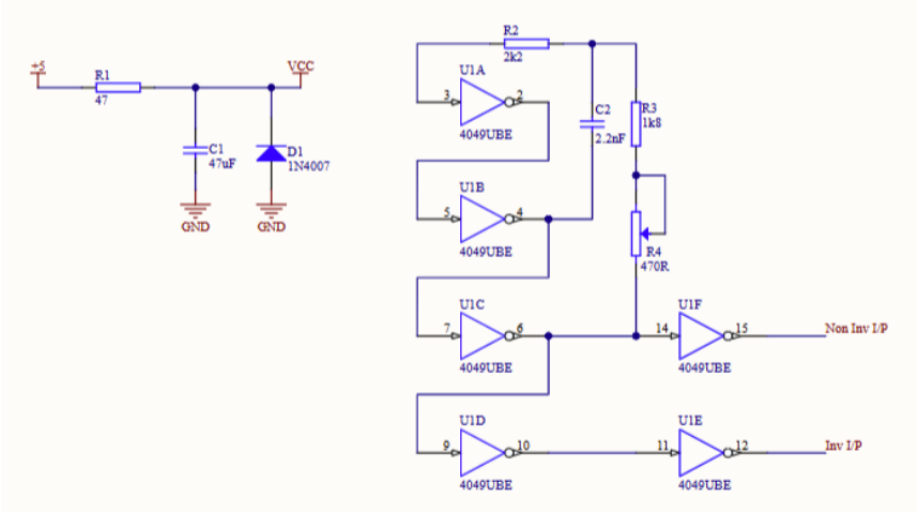


Figure 3: Oscillator Module Schematic

The Oscillator Module serves to convert the input 5V DC source into an oscillating AC square wave, as well as control the frequency of the AC source that the rest of the circuit is receiving. In the circuit diagram shown in **Figure 3**, the components UIA-UIF are representative of a CD4049UBE IC Inverter, and R4 is a 470R variable resistor. The variable resistor serves as a tuning resistor to change the frequency of the output of this module.

b) Charge Dump Modules

The next module developed for this prototype were the Charge Dump Modules. The schematic for this module is shown below in **Figure 4**.

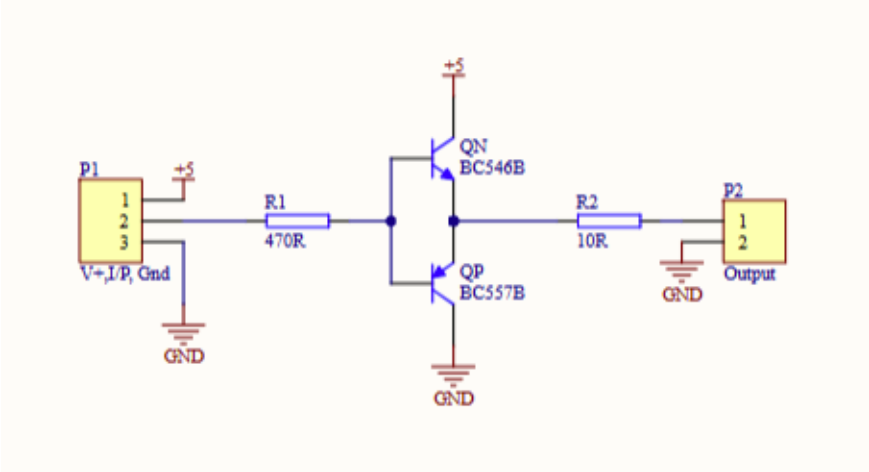


Figure 4: Charge Dump Module Schematic

The Charge Dump Modules connect to the Oscillator module, and receive the power outputted by the Oscillator. Resistor R1 is a current limiting resistor, with prevents reverse currents flowing back into the Oscillator module. QN and QP are charge dump transistors, which help to clear the stored charge from the MOSFET input capacitances in the Switching Device Modules. This improves the efficiency and reliability of the switching of the MOSFET modules. Resistor R2 is another current limiting resistor to limit the current going through the PNP transistors of the Switching Device Modules.

c) Switching Device Modules

Next, the Switching Device Modules were developed for this prototype. The schematic for these modules is shown below in **Figure 5**.

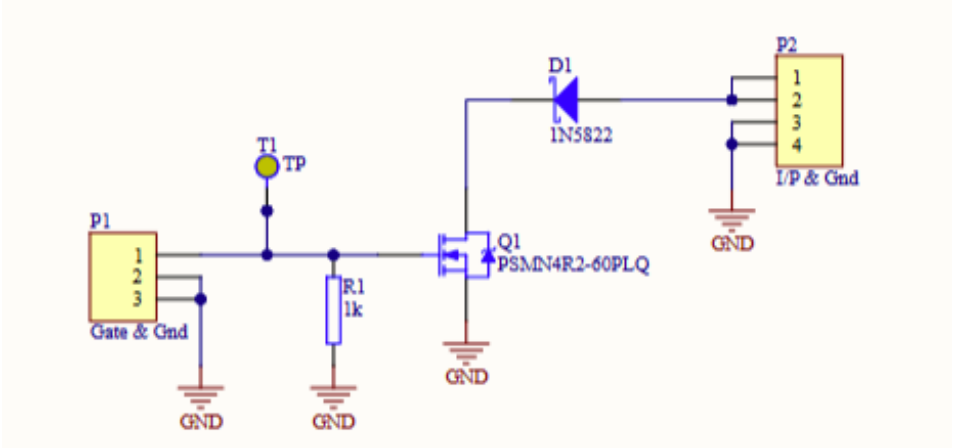


Figure 5: Switching Device Module Schematic

The Switching Device Modules serve to turn on and off depending on which phase of the input from the Oscillator they are receiving. This allows the prototype to switch its current direction through the transmitting coil, which creates the functionality of the switching currents for wireless power transfer. Resistor R1 is a gate resistor on Q1 to prevent unwanted turn-on and enhance noise immunity. It also defines the gate potential during turn-on and protects against static charge accumulation. Q1 is the MOSFET switching device which controls when the portion of the prototype with that Switching Device Module is active. The MOSFET model used for this prototype is the NDP6060L MOSFET.

d) Transformer Module

The final module constructed for this prototype is the Transformer Module. The schematic for this module is shown in **Figure 6** below.

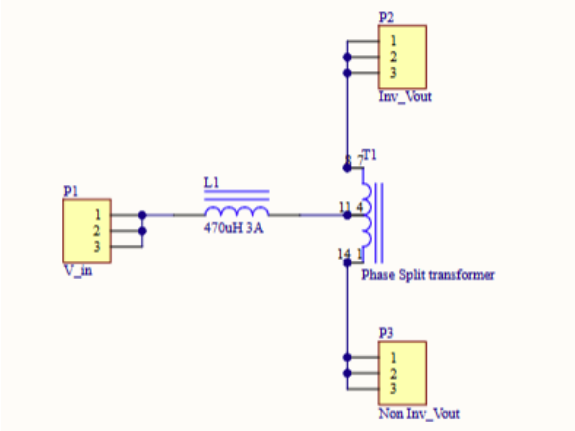


Figure 6: Transformer Module Schematic

The Transformer Module serves to filter and transform the input given by the Oscillator Module. L1 creates a low-pass filter to filter out high-order odd harmonic frequencies from the square voltage waveform, producing a sinusoidal waveform with the defined fundamental frequency from the Oscillator Module. T1 is the phase splitting transformer, which delivers power to the Switching Device Modules depending on which phase of the now sinusoidal waveform is being supplied. The inductor L1 was provided, while the transformer T1 was constructed. Using insulated copper wire with 24 turns for each side of the transformer, ferrite cores and a paper core gap to manage the saturation of the cores, the centre-tapped phase-splitting transformer was constructed. In theory, this would create a centre-tapped transformer with a total inductance of 1 mH. In practice, a centre-tapped transformer with an inductance of 0.988 mH was achieved.

e) Base Board Module

The Base Board Module, the module that connects the previously mentioned modules, was then constructed. The schematic for the Base Board Module is shown below in **Figure 7**.

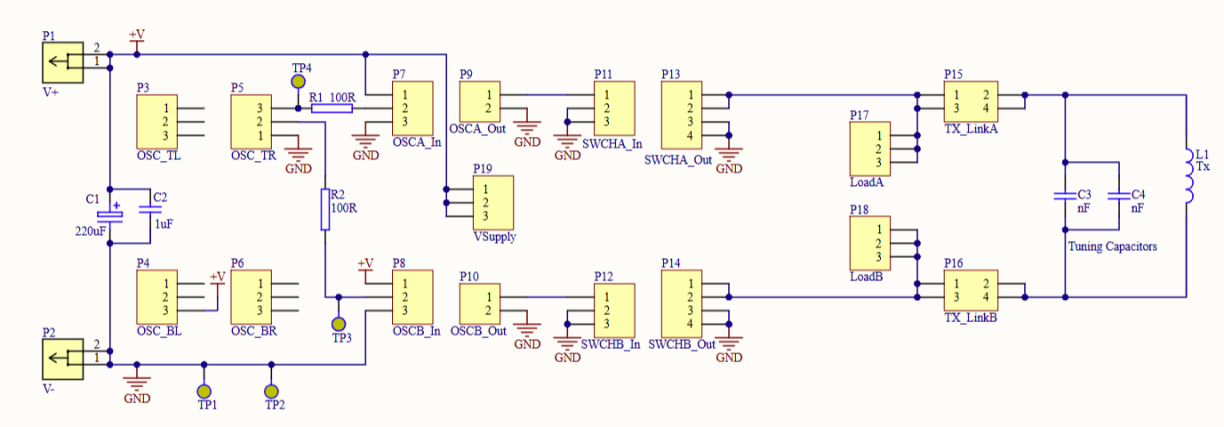


Figure 7: Base Board Module Schematic

The Base Board Module serves to connect the other modules properly, as well as add some filtering and protective components. Additionally, the Base Board Module houses the transmitting coil and the tuning capacitors to control the resonant frequency of the circuit. C1 and C2 in the schematic are EMI filtering capacitors, and C3 and C4 are the tuning capacitors to tune the transmitter resonance, used to achieve parallel resonance for optimal power transfer. L1 is the transmitting inductor coil. This coil was constructed by hand, using insulated copper wire, and contained 18 turns. This mathematically provides an inductance of about 15 µH. In practice, an inductance of about 14.8 µH was achieved.

f) Receiver Module

The final module of this prototype is the Receiver Module. The schematic for the Receiver Module is shown below in **Figure 8**.

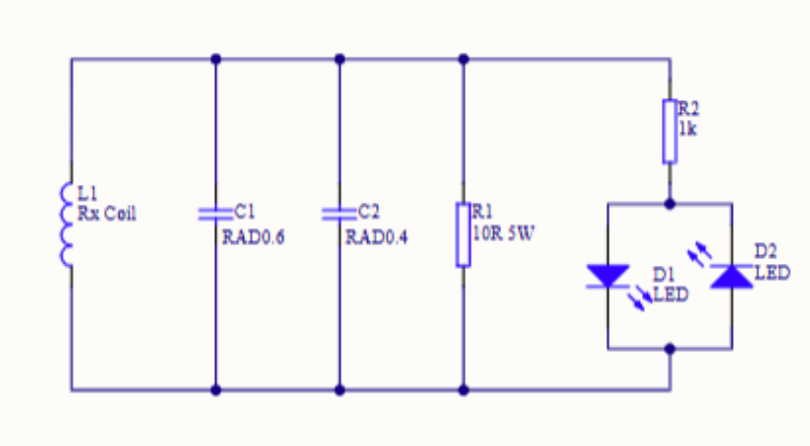


Figure 8: Receiver Module Schematic

The Receiver Module receives the power transferred wirelessly through the Base Boar Module and turns on two indicator LEDs with that receive power. Inductor L1 in the schematic is constructed in the same was as inductor L1 in the Base Board Schematic, the transmitting coil. Capacitors C1 and C2 in the schematic are tuning capacitors, used to tune the Receive Module to match the resonance of the Base Board Module. R1 is the load resistor, used to dissipate the power received, R2 is the current limiting resistor to prevent damage to the LEDs, and diodes D1 and D2 are the indicator LEDs to show power transfer.

3. Results

Shown in **Figures 9 and 10** below is the final prototype built in this lab.



Figure 9: Final WPT Prototype



Figure 10: Final Receiver Board Prototype

a) Experiment 1: Field Effect Transistors

The first experiment conducted during this lab served to test the functionality of the MOSFET switching, as well as to compare the modules used for this prototype with some other possible choices. The MOSFET modules for this prototype used the NPD6060L MOSFET, but this experiment also tested the PSMN2R5 MOSFET model.

First, a wide-range receiver, provided in the lab, is used in place of the Oscillator Module. “NO Charge Dump” modules, also provided in the lab, are used in place of the Charge Dump Modules, and a switching test load module is also used, along with the Switching Device Modules. Using a DC Power Supply of 5V, the potentiometer of the wide-range receiver module is used to vary the frequency over the full range of approximately 20kHz to 80kHz. A dual channel oscilloscope was used to measure the signals of the Gate and Drain of the FET, as well as the load resistor over that frequency range. In **Figures 11-16** below, the waveforms for these tests are shown.

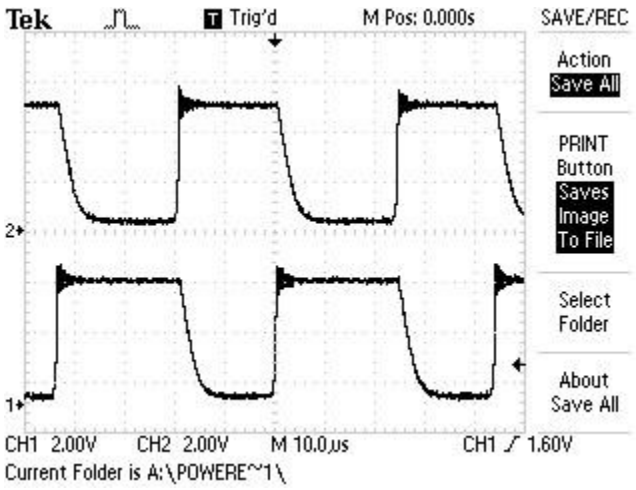


Figure 11: Waveforms of NPD6060L Drain with “NO Charge Dump” Modules at 20kHz

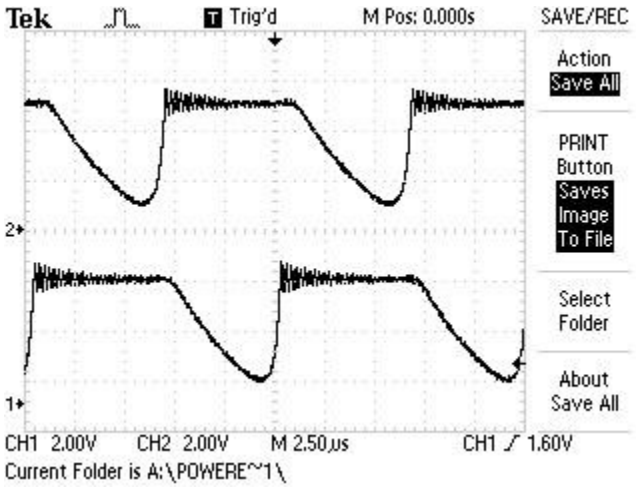


Figure 12: Waveforms of NPD6060L Drain with “NO Charge Dump” Modules at 80kHz

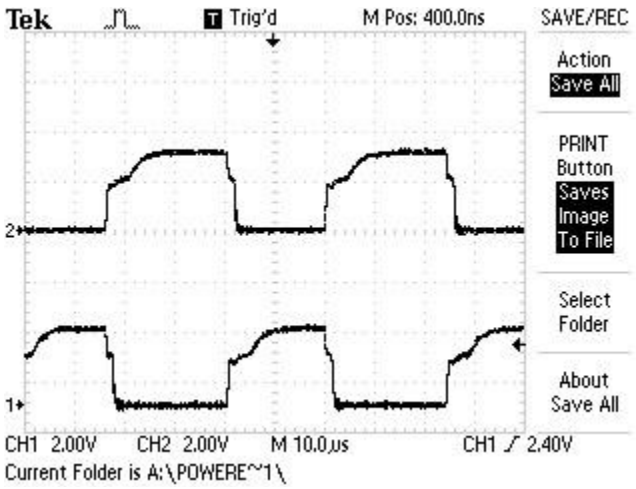


Figure 13: Waveforms of NPD6060L Gate with "NO Charge Dump" Modules at 20kHz

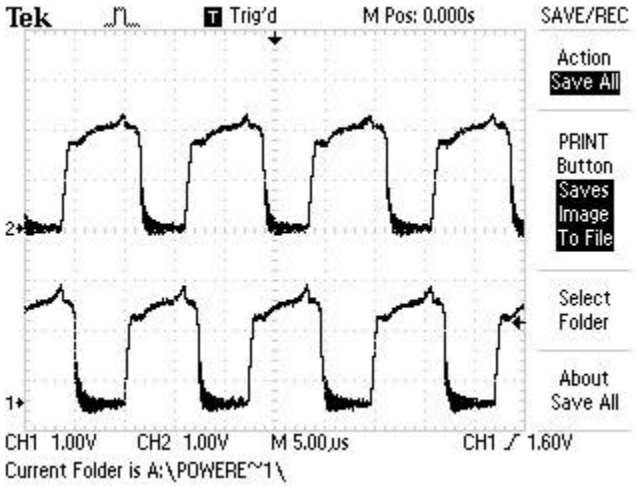


Figure 14: Waveforms of NPD6060L Gate with "NO Charge Dump" Modules at 80kHz

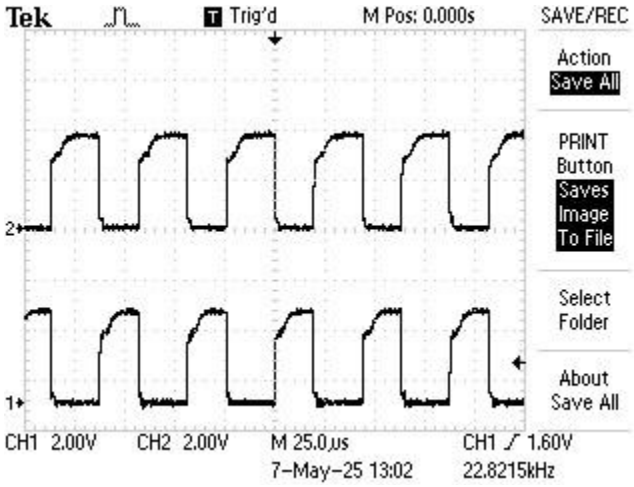


Figure 15: Waveforms of Load Resistor with "NO Charge Dump" Modules at 20kHz

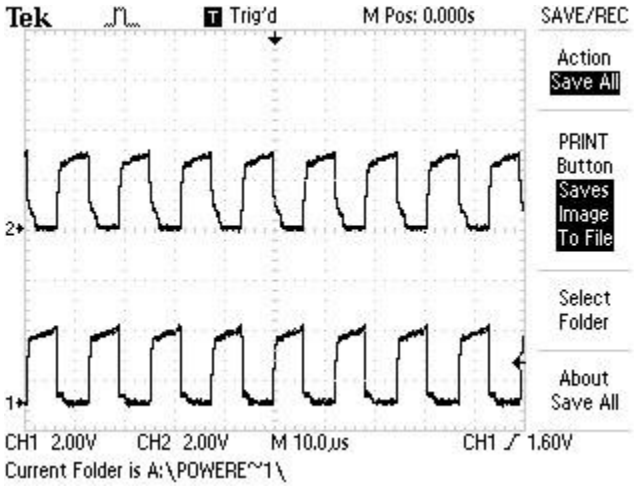


Figure 16: Waveforms of Load Resistor with "NO Charge Dump" Modules at 80kHz

From these waveforms, we can see that as the frequency of the waveforms increases, the switching speed of the devices increases as well, and the smoothness of the waveforms decreases because the switches cannot switch fast enough to compensate for the high frequency. Additionally, there were significant voltage spikes in the on-states of the MOSFETS. Without the Charge Dump Modules, the switching becomes less efficient and less reliable. It is clear that while one switching module is active, the other is inactive, and vice versa, demonstrating that one module is active on the positive cycle of the pulse from the wide-range receiver and the other module is active on the negative cycle.

Doing the same tests, but switching one of the MOSFET Switching Device Modules for the module using the PSMN2R5, we can compare these two components. **Figures 17-22** below

show the results of testing the PSMN2R5 and MNPD6060L MOSFETs, as well as the load resistors with this new model.

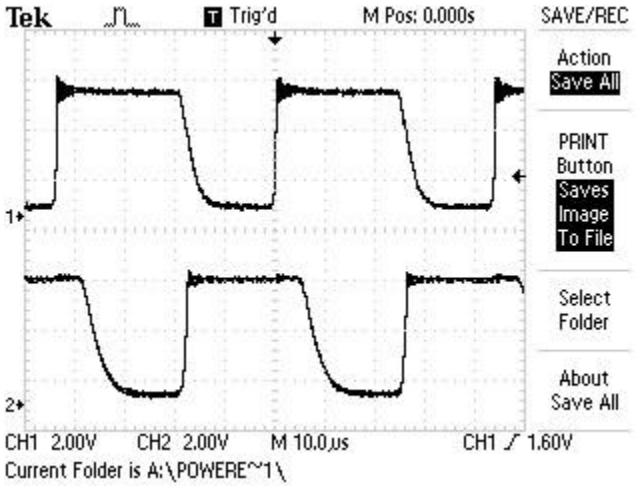


Figure 17: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Drain with "NO Charge Dump" Modules at 20kHz

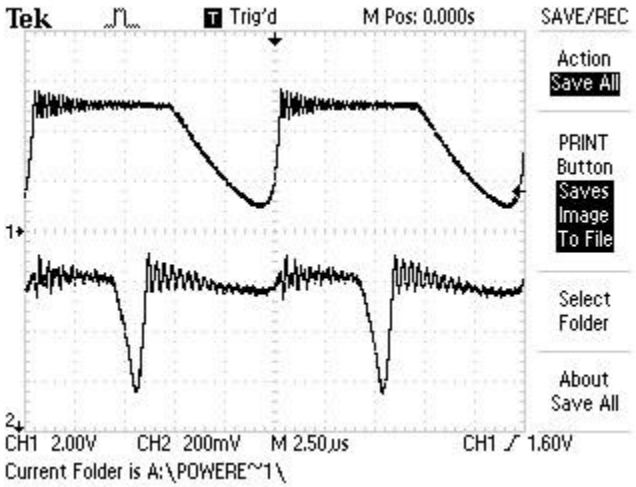


Figure 18: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Drain with "NO Charge Dump" Modules at 80kHz

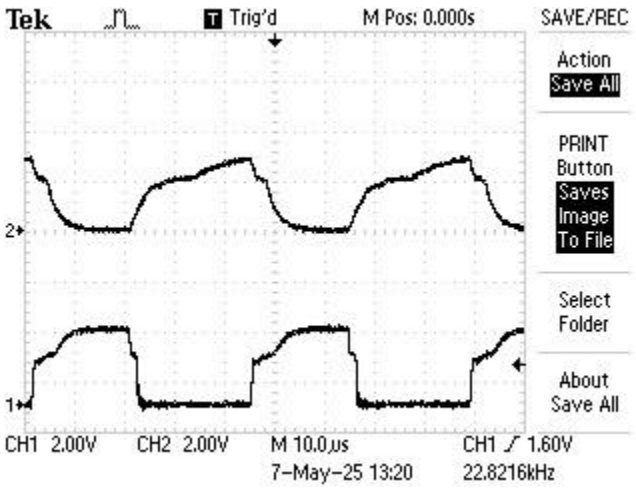


Figure 19: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Gate with "NO Charge Dump" Modules at 20kHz

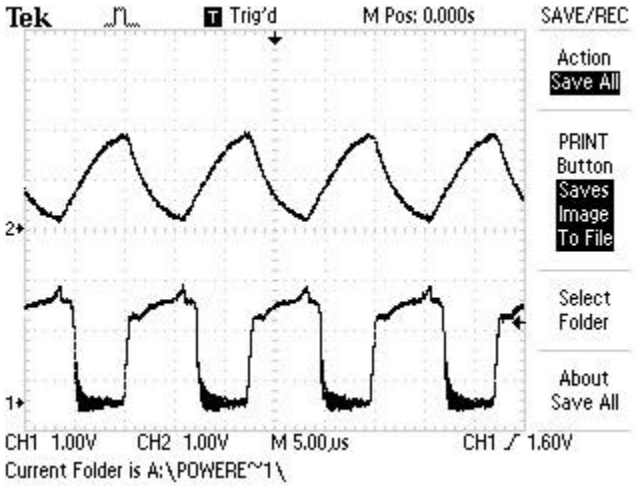


Figure 20: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Gate with "NO Charge Dump" Modules at 80kHz

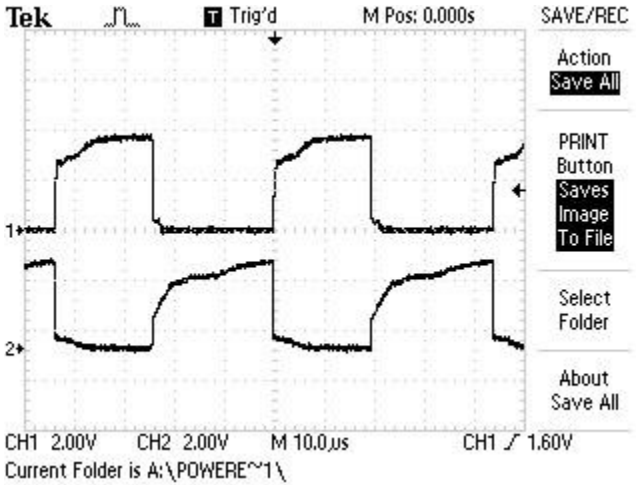


Figure 21: Waveforms of Load Resistors of NPD6060L (Channel 1) and PSMN2R5 (Channel 2) with "NO Charge Dump" Modules at 20kHz

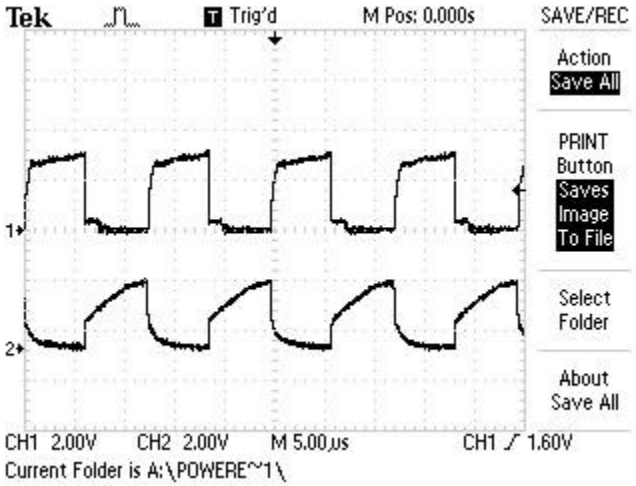


Figure 22: Waveforms of Load Resistors of NPD6060L (Channel 1) and PSMN2R5 (Channel 2) with "NO Charge Dump" Modules at 80kHz

Comparing the waveforms from the NPD6060L MOSFET and the PSMN2R5 MOSFET, there is noticeable difference. At low frequencies, the waveforms for the NPD6060L and PSMN2R5 look very similar. However, as we increase the frequency, there becomes noticeable differences. At the high frequencies, the PSMN2R5, channel 2 in **Figures 17-22**, struggles to switch fast enough to compensate for the high frequency, causing deformities in the waveforms. This is due to the difference in internal capacitances. The NPD6060L has a listed internal capacitance of 1190 pF, while the PSMN2R5 has a listed internal capacitance of 11700 pF. This much higher capacitance prevents quickly changing voltages and thus limits the switching speed of the MOSFET.

Now the Charge Dump Modules are used in place of the “NO Charge Dump” modules for both MOSFET models, and the same experiment is carried out. **Figures 23-28** show the results for these tests.

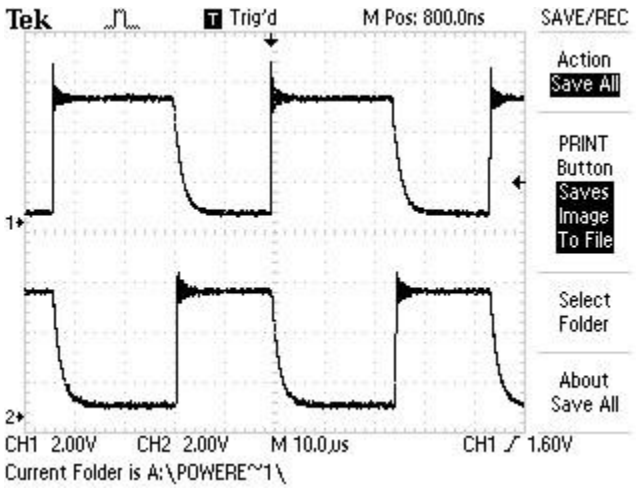


Figure 23: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Drain with Charge Dump Modules at 20kHz

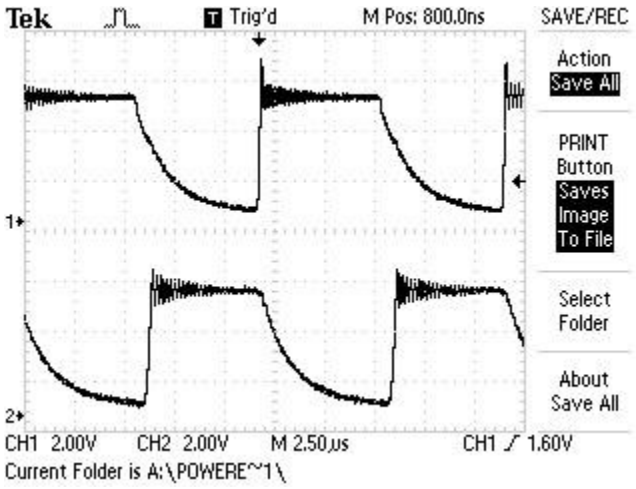


Figure 24: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Drain with Charge Dump Modules at 80kHz

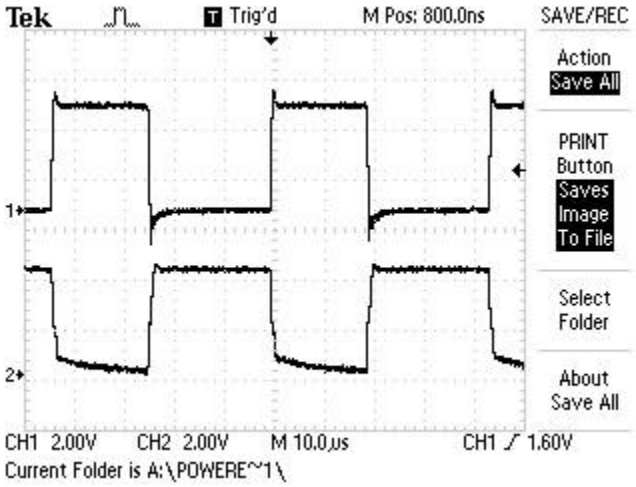


Figure 25: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Gate with Charge Dump Modules at 20kHz

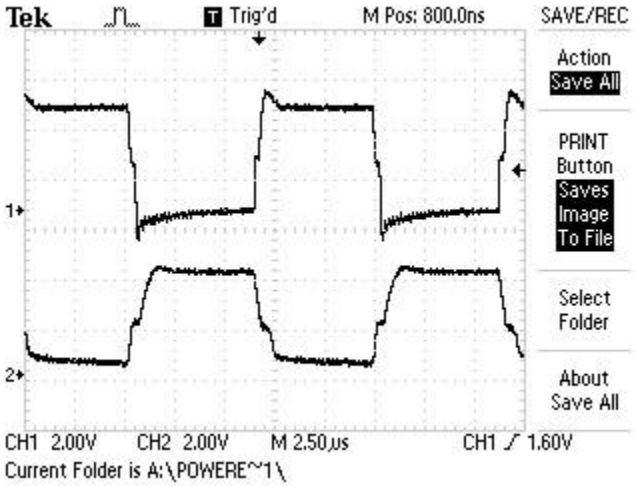


Figure 26: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Gate with Charge Dump Modules at 80kHz

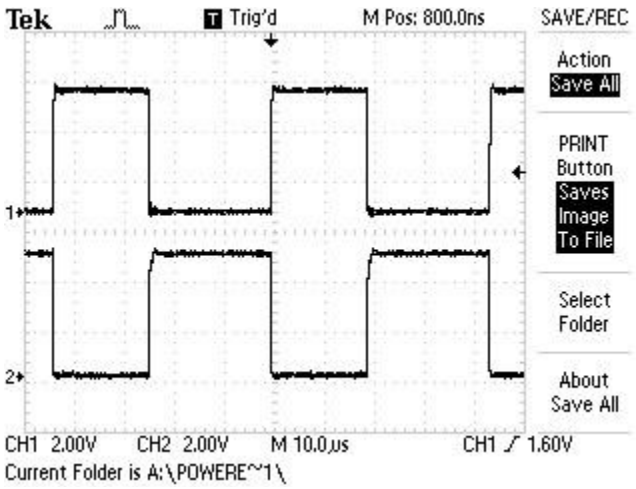


Figure 27: Waveforms of Load Resistors of NPD6060L (Channel 1) and PSMN2R5 (Channel 2) with Charge Dump Modules at 20kHz

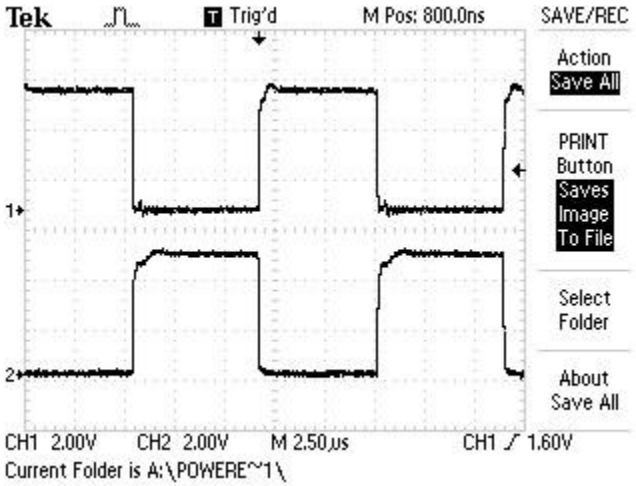


Figure 28: Waveforms of Load Resistors of NPD6060L (Channel 1) and PSMN2R5 (Channel 2) with Charge Dump Modules at 80kHz

With the Charge Dump Modules in place, it is clear that the Switching Device Modules and the load resistor are receiving a waveform that is much closer to a square wave than without the Charge Dump Modules, demonstrating the effectiveness of including these modules to improve the efficiency of the circuit. However, on the switch-on of the MOSFET, the drain experiences a more significant voltage spike. The waveforms are much closer in similarity, both between the NDP6060L model and the PSMN2R5 model, as well as between low and high frequency. The charge dump module serves to assist in discharging the capacitance of the MOSFETs, and so with these modules in place, the significant difference in internal capacitance of the MOSFETs is much less significant in the effect on the circuit.

The last part of this experiment is to replace the Charge Dump Modules with “FET Driver” modules provided in the lab and repeat the same steps again. The results for this test are shown in **Figures 29-34** below.

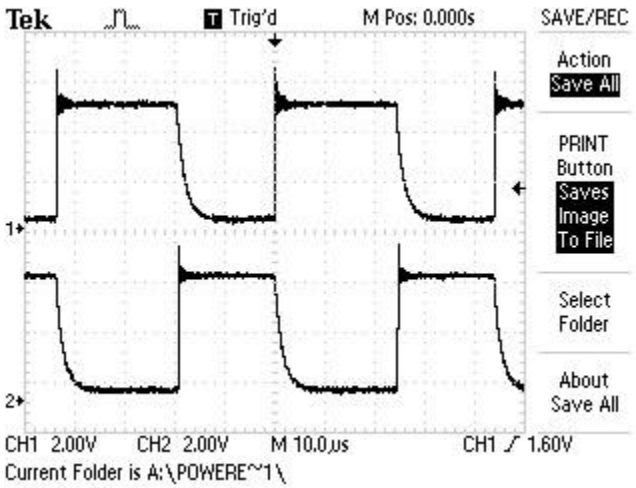


Figure 29: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Drain with “FET Driver” Modules at 20kHz

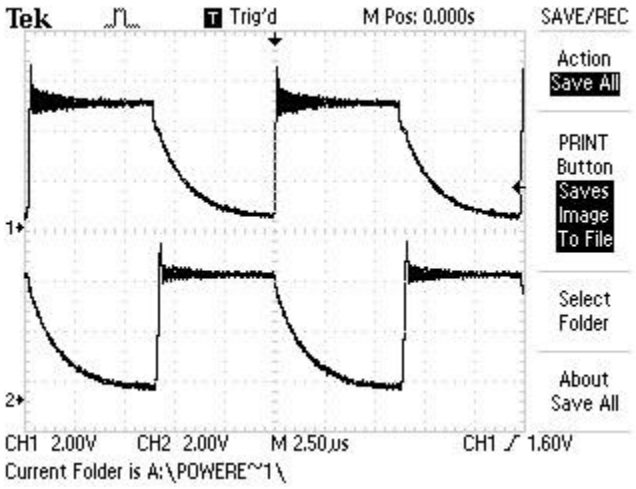


Figure 30: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Drain with “FET Driver” Modules at 80kHz

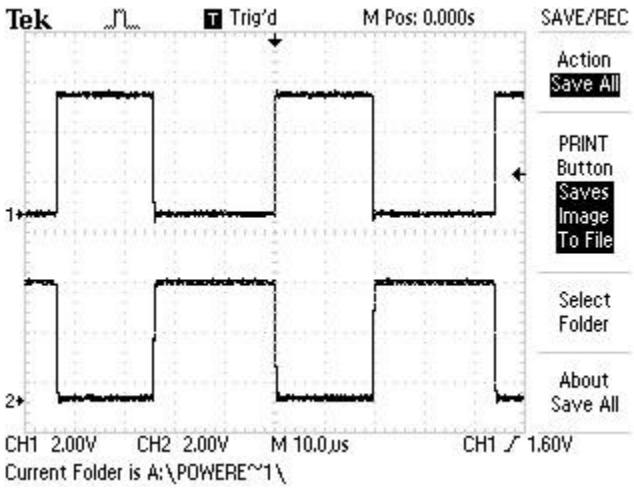


Figure 31: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Gate with "FET Driver" Modules at 20kHz

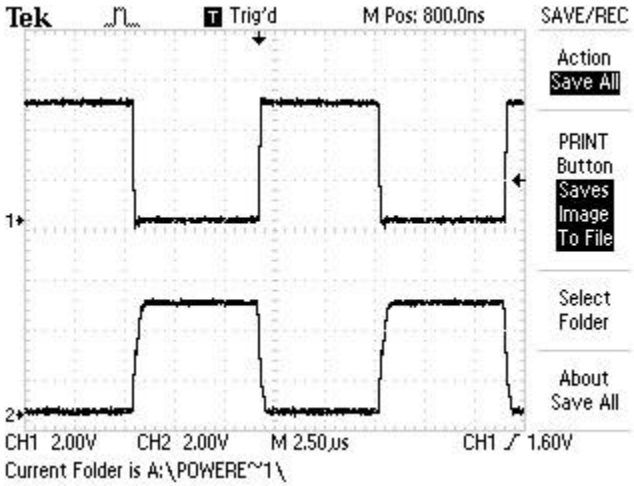


Figure 32: Waveforms of NDP6060L (Channel 1) and PSMN2R5 (Channel 2) Gate with "FET Driver" Modules at 80kHz

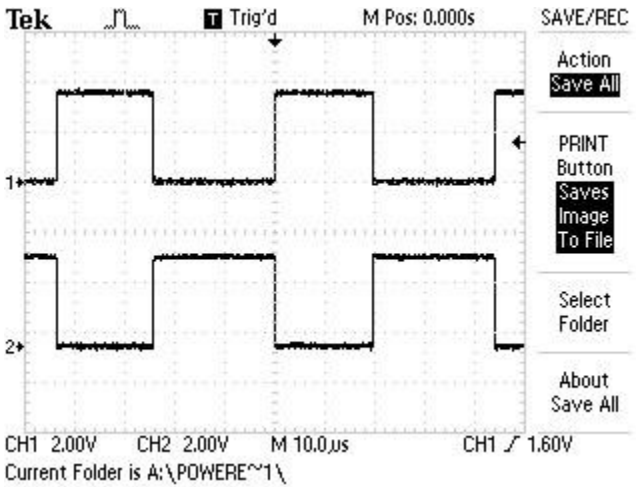


Figure 33: Waveforms of Load Resistors of NPD6060L (Channel 1) and PSMN2R5 (Channel 2) with "FET Driver" Modules at 20kHz

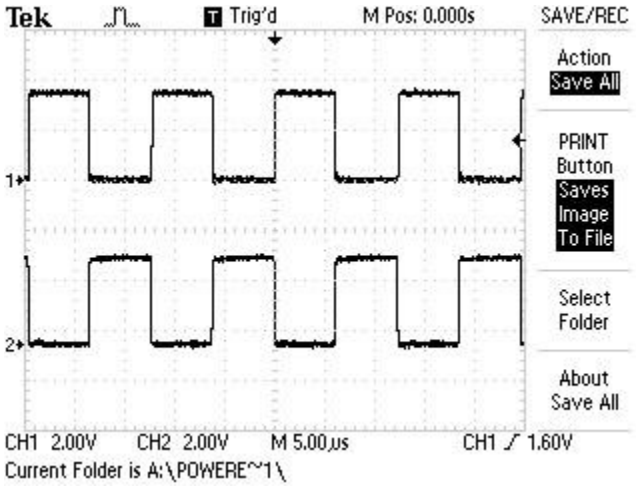


Figure 34: Waveforms of Load Resistors of NPD6060L (Channel 1) and PSMN2R5 (Channel 2) with "FET Driver" Modules at 80kHz

From this experiment, using the "FET Driver" modules provides very smooth square waves on the load resistor and the gate of the FET. However, on the drain of the FET, there is still significant transient spikes in voltage on the turn-on of the FET. These results lead to using the NPD6060L instead of the PSMN2R5, since the NPD6060L provides less transient spikes in voltage during the turn-on, even though it has slightly less smooth waveforms than the PSMN2R5, as well as faster switching speeds. Additionally, using the Charge Dump Modules provides additional smoothness to the waveforms and discharges the capacitance much quicker to increase the switching speed.

b) Experiment 2: Wireless Power Transfer Efficiency Testing

For the second experiment, the waveforms of the full prototypes will be measured to compare different spacing between the transmitting coil and the receiving coil. First, the probes of the oscilloscope are calibrated. Then, the values of the tuning capacitors are chosen. This is done by finding the desired resonant frequency of the circuit based on the range provided by the Oscillator Module and the adjustable resistor, and measuring the value of the transmitting coil, and using those values to calculate the needed capacitance value. Because of the adjustable resistor on the Oscillator Module, the frequency is able to be tuned, which allows an appropriate range of values to be found rather than only one value. Using these methods, resonant conditions for the full circuit can be found. For this prototype, a resonant frequency of 91 kHz was used, with a transmitting inductance of 14.8 µH and a capacitance of approximately 206 nF. Using the below equation, the resonant frequency can be confirmed.

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(14.8\mu H * 206nF}} = 91{,}149.79\ Hz = 91\ kHz$$

Resonant Frequency Calculation

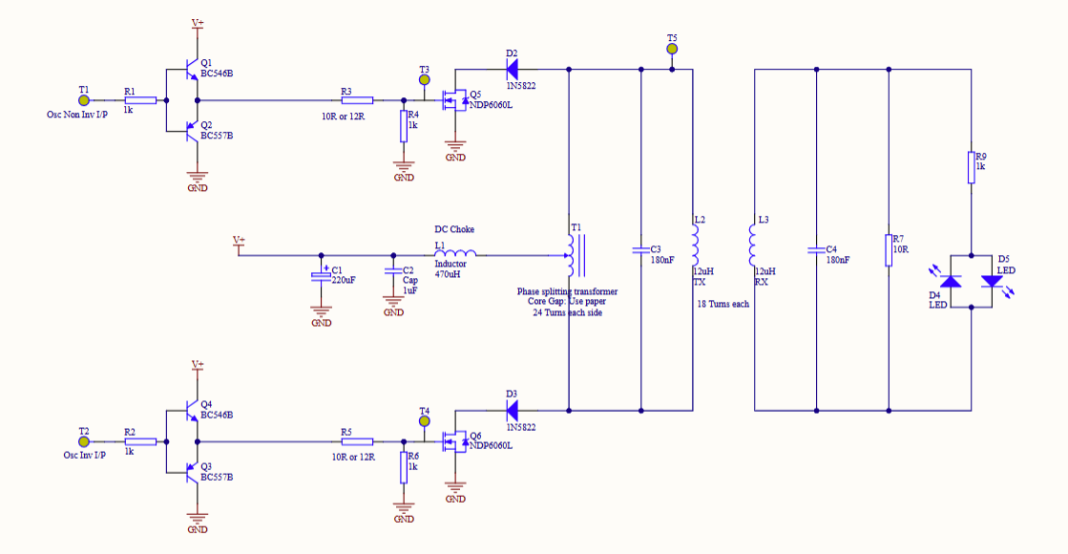
Once the resonant conditions of the circuit are found, one can compare the power transfer efficiency of the circuit. First, the circuit is tuned to the resonant conditions, and the idle current of the Base Board is measured. For this prototype, an idle current of 0.156 A was achieved. Next, a series of spacers of varying thickness are inserted between the transmitting coil and the receiving coil, and the efficiencies of power transfer are calculated. Spacer thicknesses of 2mm, 3mm, 4.5mm, 6mm, and 10mm were tested. Knowing the resistance of the load resistor to be 10 Ohms, the efficiency of the circuit can be calculated by finding the input and output power. The results of the tests are shown in **Table 1** below.

Table 1: Efficiency of Prototype Board with Varying Spacer Thicknesses

Thickness	2mm	3mm	4.5mm	6mm	10mm
Input Voltage (V)	4.85	4.85	4.85	4.85	4.85
Input Current (A)	1.968 (overload)	1.846	1.223	1.051	0.623
Output Voltage (V RMS)	N/A	7.53	6.15	5.70	4.21
Input Power (W)	N/A	8.9531	5.9193	5.0868	3.0153
Output Power (W)	N/A	5.6701	3.7823	3.2490	1.7724
Efficiency (%)	N/A	63.33	63.90	63.87	58.78

These results show that the highest efficiency is achieved when a spacing of 4.5mm is used between the transmitting coil and the receiving coil, with an efficiency of 63.9%.

4. Discussion



Complete Schematic Excluding Oscillator Module

During this lab, no major anomalies or issues arose. During construction, there were testing complications with power not properly being transferred to the transmitting coil. However, this was easily resolved once the issue was found. The issue in this case was that the insulation on the copper wire of the inductor was not properly removed, and so the wires did not have proper connection to the pads of the Base Board. Once this was resolved, the power transfer was fully functional.

Additionally, the frequency range provided by the adjustable resistor on the Oscillator Module was lower than expected. For this prototype, the range was approximately 60kHz – 98 kHz, whereas the expected range was approximately 80kHz – 110 kHz. This did not end up being a significant issue however, as the tuning capacitors on the Base Board are able to compensate for this. Having a slightly higher capacitance lowers the resonant frequency, and by increasing the tuning capacitance, the resonant frequency became within the range that the Oscillator Module covered.

5. Conclusion

Through this lab, fundamental knowledge on Wireless Power Transfer was gained, as well as knowledge on the construction and testing of wireless power transfer modules and circuits in general. Additionally, knowledge on efficiency and tuning was gained, and the functionality of each of the modules was understood.